

Perplexity

A Comprehensive Research Biography

14 July 2026

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Chapter 1

Introduction

In information theory, perplexity is a measure of uncertainty for a discrete probability distribution. The perplexity of a fair coin toss is 2, and that of a fair die roll is 6; and generally, for a probability distribution with exactly N outcomes each having a probability of exactly $1 / N$, the perplexity is simply N . But perplexity can also be applied to unfair dice, and to other non-uniform probability distributions. It can be defined as the exponentiation of the distribution's information entropy. Intuitively, the larger the perplexity, the more "surprised" is an observer by a value drawn from the distribution, on average. In ecology, the perplexity is also referred to as the (order-1 true) diversity. In statistical modeling and machine learning, perplexity is also used to measure how well a proposed probability model matches a given unknown probability distribution. It is then defined as the exponentiation of the cross entropy of the two distributions. The smaller the perplexity, the better the model. Perplexity was originally introduced in 1977 in the context of speech recognition by Frederick Jelinek, Robert Leroy Mercer, Lalit R. Bahl, and James K. Baker.

Chapter 3

Career and Contributions

Perplexity of a probability distribution

The perplexity PP of a discrete probability distribution p is defined as $PP(p) = \prod_x p(x)^{-p(x)}$, $\{\mathrm{PP}(p) = \prod_x p(x)^{-p(x)}\}$ where $p(x)$ denotes the probability of the outcome x , the product ranges over all possible outcomes, and the convention $0 \cdot 0 = 1$ is used. This can also be written as $PP(p) = b^{-\sum_x p(x) \log_b p(x)}$, $\{\mathrm{PP}(p) = b^{-\sum_x p(x) \log_b p(x)}\}$ where the value of b does not affect the result (e.g., b can be chosen to be 2, 10, e , or any other positive value other than 1) and the convention $0 \log_b 0 = 0$ is used. The logarithm $\log PP(p)$ is the entropy of the distribution; it is given the unit shannon (bit) if the base of the logarithm is 2, and nat if the natural logarithm is used. Perplexity of a random variable X may be defined as the perplexity of the distribution over its possible values x . It can be thought of as a measure of uncertainty or "surprise" related to the outcomes. For a probability distribution p where exactly k outcomes each have a probability of $1/k$ and all other outcomes have a probability of zero, the perplexity of this distribution is simply k . This is because the distribution models a fair k -sided die, with each of the k outcomes being equally likely. In this context, the perplexity k indicates that there is as much uncertainty as there would be when rolling a fair k -sided die. Even if a random variable has more than k possible outcomes, the perplexity will still be k if the distribution is uniform over k outcomes and zero for the rest. Thus, a random variable with a perplexity of k can be described as being "k-ways perplexed", meaning it has the same level of uncertainty as a fair k -sided die. Perplexity is sometimes used as a measure of the difficulty of a prediction problem. It is, however, generally not a straightforward representation of the relevant probability. For example, if you have two choices, one with probability 0.9, your probability of a correct guess using the optimal strategy is 0.9. Yet, the perplexity is $0.9 \cdot 0.9 \cdot 0.1 \cdot 0.1 = 1.38$. The inverse of the perplexity, $1/1.38 = 0.72$, is not equal to the probability 0.9. The perplexity is the exponentiation of the entropy, a more commonly encountered quantity. Entropy measures the expected or "average" number of bits required to encode the outcome of the random variable using an optimal variable-length code. It can also be regarded as the expected information gain from learning the outcome of the random variable, providing insight into the uncertainty and complexity of the underlying probability distribution.

Perplexity of a probability model

A model of an unknown probability distribution p may be proposed based on a training sample that was drawn from p . Given a proposed probability model q , one may evaluate q by asking how well it predicts a separate test sample $x_1, x_2, \dots, x_N$ also drawn from p . The perplexity of the model q is defined as $b^{-1/N \sum_{i=1}^N \log_b q(x_i)} = (\prod_{i=1}^N q(x_i))^{-1/N}$, $\{\mathrm{perplexity}(q) = b^{-1/N \sum_{i=1}^N \log_b q(x_i)} = (\prod_{i=1}^N q(x_i))^{-1/N}\}$

$$b^{-\frac{1}{N} \sum_{i=1}^N \log_b q(x_i)} = \left(\prod_{i=1}^N q(x_i) \right)^{-1/N},$$
 where b is customarily 2. Better models q of the unknown distribution p will tend to assign higher probabilities $q(x_i)$ to the test events. Thus, they have lower perplexity because they are less surprised by the test sample. This is equivalent to saying that better models have higher likelihoods for the test data, which leads to a lower perplexity value. The exponent, $-1/N \sum_{i=1}^N \log_b q(x_i)$, with $b = 2$, may be regarded as the average number of bits needed to represent a test event x_i if one uses an optimal code based on q . Low-perplexity models do a better job of compressing the test sample, requiring few bits per test element on average because $q(x_i)$ tends to be high. The exponent may also be interpreted as a cross-entropy: $H(\tilde{p}, q) = -\sum_x \tilde{p}(x) \log_b q(x)$, where $\tilde{p}$ denotes the empirical distribution of the test sample (i.e., $\tilde{p}(x) = n/N$ if x appeared n times in the test sample of size N). By the definition of KL divergence, it is also equal to $H(\tilde{p}) + \text{DKL}(\tilde{p} \parallel q)$, which is $\geq H(\tilde{p})$. Consequently, the perplexity is minimized when $q = \tilde{p}$.

Token-normalized perplexity

In natural language processing (NLP), a corpus is a structured collection of texts or documents, and a language model is a probability distribution over entire texts or documents. Consequently, in NLP, the more commonly used measure is token-normalized perplexity (a token being a word or, more frequently, sub-word), defined as:
$$\left(\prod_{i=1}^n q(s_i) \right)^{-1/N}$$
 where $s_1, \dots, s_n$ are the n documents in the corpus and N is the number of tokens in the corpus. This normalizes the perplexity by the length of the text, allowing for more meaningful comparisons between different texts or models rather than documents. Suppose the average text x_i in the corpus has a probability of 2^{-190} according to the language model. This would give a model perplexity of 2^{190} for a sentence. However, in NLP, it is more common to normalize by the length of a text. Thus, if the test sample has a length of 1000 tokens, and could be coded using 7.95 bits per token, one could report a model perplexity of token of $2^{7.95} = 247$. In other words, the model is as confused on test data as if it had to choose uniformly and independently among 247 possibilities for each token. There are two standard evaluation metrics for language models: perplexity or word error rate (WER). The simpler of these measures, WER, is simply the percentage of erroneously recognized words E (deletions, insertions, substitutions) to total number of words N , in a speech recognition task i.e.
$$\text{WER} = \left(\frac{E}{N} \right)$$
 The second metric, perplexity (of a token), is an information theoretic measure that evaluates the similarity of proposed model m to the original distribution p . It can be computed as a inverse of (geometric) average probability of test set T
$$\text{PPL}(D) = \frac{1}{m(T)^N} = 2^{-\frac{1}{N} \log_2(m(T))},$$
 where N is the number of tokens in test set T . This equation can be seen as the exponentiated cross entropy, where cross entropy $H(p; m)$ is approximated as
$$H(p; m) = -\frac{1}{N} \log_2(m(T)).$$

$$H(p;m)=-\{\frac{1}{N}\}\log _2\left(m(T)\right).$$

Recent advances in language modeling

Since 2007, significant advancements in language modeling have emerged, particularly with the advent of deep learning techniques. Token-normalized perplexity, a measure that quantifies the predictive power of a language model, has remained central to evaluating models such as the dominant transformer models like Google's BERT, OpenAI's GPT-4 and other large language models (LLMs). This measure was employed to compare different models on the same dataset and guide the optimization of hyperparameters, although it has been found sensitive to factors such as linguistic features and sentence length. Despite its pivotal role in language model development, perplexity has shown limitations, particularly as an inadequate predictor of speech recognition performance, overfitting and generalization, raising questions about the benefits of blindly optimizing perplexity alone.

Brown Corpus

The lowest perplexity that had been published on the Brown Corpus (1 million words of American English of varying topics and genres) as of 1992 is indeed about 247 per word/token, corresponding to a cross-entropy of $\log_2 247 = 7.95$ bits per word or 1.75 bits per letter using a trigram model. While this figure represented the state of the art (SOTA) at the time, advancements in techniques such as deep learning have led to significant improvements in perplexity on other benchmarks, such as the One Billion Word Benchmark. In the context of the Brown Corpus, simply guessing that the next word is "the" will achieve an accuracy of 7%, contrasting with the $1/247 = 0.4\%$ that might be expected from a naive use of perplexity. This difference underscores the importance of the statistical model used and the nuanced nature of perplexity as a measure of predictiveness. The guess is based on unigram statistics, not on the trigram statistics that yielded the perplexity of 247, and utilizing trigram statistics would further refine the prediction.

See also

Cross-entropy Statistical model validation == References ==

Chapter 4

Recent Developments and Current Work

Nvidia Vs. AMD: Perplexity Choosing Nvidia Over AMD Tells a Deeper Story About Chip Dominance - Yahoo Finance

Yahoo Finance | 2026-07-12

NVDA's \$82B quarter dwarfs AMD's \$10B as Blackwell, NVLink, and CUDA create a unified stack AMD is still racing to match. Perplexity chose NVIDIA's Vera CPUs over x86 chips, with the Vera CPUs running 1.5x faster, a move that signals NVIDIA is now threatening AMD's EPYC server CPU stronghold directly. AMD's 141% YTD run and 184x trailing P/E leave no margin for error when its MI450 volume ramp lands in H2 2026. Don't wait: the analyst who called NVIDIA in 2010 just revealed his top 10 AI stocks. See the full list FREE now . NVIDIA (NASDAQ: NVDA) and AMD (NASDAQ: AMD) both closed strong quarters, but the ground shifted this week when Perplexity picked NVIDIA's new Vera CPUs over x86 server chips for its multi-agent AI coding stack, running 1.5 times faster than standard server processors. That decision reframes the earnings comparison. One company now sells GPUs, CPUs, networking, and software as one bundle. The other is still assembling its answer. NVIDIA's Q1 FY27 showed revenue of \$81.61B, up 85.2% YoY, with Data Center at \$75.25B (+92%) and Networking at \$14.8B (+199%). Jensen Huang framed it plainly: "The buildout of AI factories, the largest infrastructure expansion in human history, is accelerating at extraordinary speed." Blackwell, NVLink Fusion, and Spectrum-X are selling as one unified stack. AMD's Q1 FY26 was solid but smaller. Revenue hit \$10.25B (+37.9% YoY), Data Center reached \$5.78B (+57%), and non-GAAP EPS came in at \$1.37. Lisa Su leaned on the pipeline: "Customer engagement around MI450 Series and Helios is strengthening, with leading customer forecasts exceeding our initial expectations." The MI450 volume ramp lands in H2 2026. The Perplexity win validates NVIDIA's push into a new \$20 billion CPU vertical that used to belong to x86 vendors. NVIDIA is parlaying GPU share into host-CPU sockets, exactly the margin territory AMD's EPYC franchise defends. AMD fights a two-front war: chase NVIDIA in accelerators with MI450/Helios and protect server CPU turf from Vera. Customer deals look close on paper (OpenAI signed 10+ GW with NVIDIA and 6 GW with AMD), but NVIDIA's CUDA-X and Dynamo software layer keeps inference workloads sticky. ROCm 7 is improving but still short of moat status. First, whether MI450 shipments in H2 2026 meet the "exceeding expectations" language Su used, since AMD's 184x trailing P/E leaves no room for a slip. Second, whether Vera CPU adoption spreads beyond Perplexity into hyperscalers. NVIDIA guided Q2 revenue to \$91.0B, and analysts carry a target of \$301.62. I lean NVIDIA here. The Perplexity decision signals that inference buyers default to Blackwell plus NVLink plus CUDA when latency matters. That is a moat. AMD remains healthy. Data Center up 57% and free cash flow of \$2.57B (+253%) show the business is compounding. But shares already ran 140.99% year

to date, and last week gave back 11.15%. For the platform winner, NVIDIA is cleaner. For the higher-variance catch-up trade, AMD works, provided Helios ships on time.

Gemini, Perplexity and ChatGPT have picked their World Cup winner – and it's unanimous - PC Guide

PC Guide | 2026-07-13

The World Cup is drawing to a close, and only four teams remain. We know that anything can happen in a soccer game, but is there enough data to suggest who will win the tournament? With the France vs Spain and England vs Argentina semi-finals to come, who would leading AI chatbots pick as the likely champs come final day, July 19? The question was simple: Which team do you think wins the 2026 World Cup? Predicting a definite winner is incredibly tough right now because the remaining field consists entirely of the tournament's pre-seeded top four heavyweights. Anyone left has the quality to lift the trophy: If you are looking for the safest bet, France (28%) is the smartest choice given their ruthless consistency and squad depth. But if Spain can dictate the tempo in their semi-final match, the entire field opens up. That's one pick for France then. It is somewhat surprising to see Argentina at the bottom here. Yet they have had to fight back from near elimination by Egypt. Plus, in the quarter-final against Switzerland, the match was close until the Swiss team was reduced to ten men through a sending-off. If I had to pick one team, I'd say France. They're consistently near the top of the strongest pre-tournament models and expert picks, with one major supercomputer prediction giving France the highest title chance at 18.66%, ahead of Argentina and Spain. Why France France keeps showing up as the safest overall bet because it combines elite talent, depth, and recent tournament pedigree, which is why multiple previews and odds boards place it at or near the top. Close challengers Spain is the other team I'd take very seriously, and in some models it is the top favorite or only a step behind France. France is the standout option for Perplexity, too, thanks to the team's strong talent pool, tournament data, and odds-on status. Perplexity also opts for Spain as the other team to take seriously, which, as a soccer fan, I understand. However, Spain would need to beat France... so, that seems a little obvious in terms of momentum. Based on current form, expert analysis, and statistical models, France are the slight favorites to win the tournament. They have been highly consistent throughout the competition, and several prediction models, including Opta's supercomputer, give them the best chance of lifting the trophy. That said, with France, Spain, England, and Argentina all remaining, the margins are very small. Any of those four teams are capable of winning the World Cup, and a single moment—a red card, penalty shootout, or individual brilliance—could decide the tournament. So a third pick for France. England fans may be buoyed (or not) to see ChatGPT rating the Three Lions highly, but again, I'm somewhat surprised to see Argentina left last. ChatGPT does note that “a red card, penalty shootout, or individual brilliance” could decide the tournament. And the latter of those could easily apply to Argentina's Lionel Messi if Argentina makes it to the final. England's Jude Bellingham has proven

capable of creating magic, too. Of course, anything can happen, but it seems these three AI tools are in agreement. And they share that view with Opta's supercomputer, which offers up France (33.58%), Spain (23.84%), England (22.62%), and Argentina (19.96%). At least it's not long until we know for sure. Kevin has a broad interest and enthusiasm for consumer electronics, PCs, and all things consumer tech. And more than 17 years experience in tech journalism.

Nvidia Vs. AMD: Perplexity Choosing Nvidia Over AMD Tells a Deeper Story About Chip Dominance - 24/7 Wall St.

24/7 Wall St. | 2026-07-12

NVIDIA (NASDAQ: NVDA | NVDA Price Prediction) and AMD (NASDAQ: AMD) both closed strong quarters, but the ground shifted this week when Perplexity picked NVIDIA's new Vera CPUs over x86 server chips for its multi-agent AI coding stack, running 1.5 times faster than standard server processors. That decision reframes the earnings comparison. One company now sells GPUs, CPUs, networking, and software as one bundle. The other is still assembling its answer. NVIDIA's Q1 FY27 showed revenue of \$81.61B, up 85.2% YoY, with Data Center at \$75.25B (+92%) and Networking at \$14.8B (+199%). Jensen Huang framed it plainly: "The buildout of AI factories, the largest infrastructure expansion in human history, is accelerating at extraordinary speed." Blackwell, NVLink Fusion, and Spectrum-X are selling as one unified stack. AMD's Q1 FY26 was solid but smaller. Revenue hit \$10.25B (+37.9% YoY), Data Center reached \$5.78B (+57%), and non-GAAP EPS came in at \$1.37. Lisa Su leaned on the pipeline: "Customer engagement around MI450 Series and Helios is strengthening, with leading customer forecasts exceeding our initial expectations." The MI450 volume ramp lands in H2 2026. The Perplexity win validates NVIDIA's push into a new \$20 billion CPU vertical that used to belong to x86 vendors. NVIDIA is parlaying GPU share into host-CPU sockets, exactly the margin territory AMD's EPYC franchise defends. AMD fights a two-front war: chase NVIDIA in accelerators with MI450/Helios and protect server CPU turf from Vera. Customer deals look close on paper (OpenAI signed 10+ GW with NVIDIA and 6 GW with AMD), but NVIDIA's CUDA-X and Dynamo software layer keeps inference workloads sticky. ROCm 7 is improving but still short of moat status. First, whether MI450 shipments in H2 2026 meet the "exceeding expectations" language Su used, since AMD's 184x trailing P/E leaves no room for a slip. Second, whether Vera CPU adoption spreads beyond Perplexity into hyperscalers. NVIDIA guided Q2 revenue to \$91.0B, and analysts carry a target of \$301.62. I lean NVIDIA here. The Perplexity decision signals that inference buyers default to Blackwell plus NVLink plus CUDA when latency matters. That is a moat. AMD remains healthy. Data Center up 57% and free cash flow of \$2.57B (+253%) show the business is compounding. But shares already ran 140.99% year to date, and last week gave back 11.15%. For the platform winner, NVIDIA is cleaner. For the higher-variance catch-up trade, AMD works, provided Helios ships on time.

I asked ChatGPT, Perplexity, and Gemini to build me a gaming PC, and one made a big mistake - PC Guide

PC Guide | 2026-07-13

AI is supposed to help make our lives easier. That includes shopping and providing relevant information on potential purchases. But can it help save money on sourcing components and reliably support the planning of a PC build? Not always. As a believer in component bundles, I tasked ChatGPT, Perplexity, and Gemini with finding some before filling in the gaps. Here's what happened. My prompt to all three AI assistants was the same. I wanted to focus on a pretty powerful mid-range system build, and set a \$2,000 budget: "I'm trying to build a new gaming PC for up to \$2000. Are you able to recommend any component bundles worth looking at, including a CPU and motherboard, from US retailers" Clear enough, so I hit enter on each and let the tools do their best. The results? Pretty solid to start, with the caveat that you should definitely always check and compare prices. We saw ChatGPT specifically get pricing wrong when asking it to outbuild the Steam Machine. Prices can change quickly and are higher in 2026, so always make sure and don't take an AI tool's word on it. For a \$2,000 gaming PC, I'd focus on bundles that maximize the value of the CPU + motherboard + RAM, then spend as much as possible on the GPU. This would be my first choice if it's available at your local Micro Center: Bundle price: \$829.99 Regular combined price: \$1,099.97 Savings: About \$270 That's a comprehensive response, and it includes the best CPU for gaming. Assuming availability, the savings are solid, and the inclusion of Newegg and Micro Center offers some options for further research. Plus, I was given an immediate estimated total, so we know roughly where we are before tax. Yes – the strongest US-retailer bundle option for a \$2000 gaming PC is Micro Center's AMD CPU + motherboard + RAM combo, especially the Ryzen 7 7800X3D bundle. Micro Center also has a broader AMD bundle page, while Newegg and Best Buy have bundle/package sections worth checking if you want alternatives. Using a \$699.99 RX 7900 XT-class card and a \$120 case, the total comes to about \$1,660.96 before tax. If the GPU is closer to \$799.99 and the case is \$140, the total is about \$1,780.96 before tax Perplexity tried to offer options across retailers but, weirdly, only specified a Micro Center bundle: the Ryzen 7 7800X3D + AM5 motherboard + 32GB DDR5. Not too shabby, as the Ryzen 7 7800X3D is a fine option even if it's last-gen. I also got extra information about the full spend, depending on GPU and case costs. But it doesn't go as far as ChatGPT in completing the parts list. Gemini focused on Micro Center but gave the most detailed response of all, offering up three distinct options. I've summarized those below because they are extensive with a capital E, even including an Intel 'hybrid' option. Top marks for giving users options, Gemini. Micro Center's CPU, motherboard, and RAM combinations are a fantastic way to save cash on a \$2,000 budget, allowing you to maximize your remaining funds for a top-tier graphics card. The most compelling bundle configurations at Micro Center that align perfectly with your budget include the following: Price: ~\$699.99 (Saves roughly \$370 compared to buying standalone parts) ChatGPT was the only AI tool to have replied immediately with a full,

recommended build list. So, with a bit of prompting to complete the checklist in the others, here are the three for comparison. Although there is one specific issue here that any PC builder should spot immediately: These are all completely viable builds and, as you'd expect, each AI tool gave a little nod to the cost before tax. However, Gemini went a step further, offering an additional thought on the remaining budget. Where to Use the Leftover ~\$120: This setup leaves you comfortably under your absolute \$2,000 ceiling. You can hold onto the extra cash to absorb local state sales tax, or reallocate it toward aesthetics—like upgrading the air cooler to a premium Liquid AIO (such as the Arctic Liquid Freezer III 360) to completely fill out the top of the Montech case. Decent advice, and honestly, it's somewhat odd that Gemini should mention a CPU cooler. Somehow, inexplicably, that's precisely what Perplexity completely failed to include in the parts table or anywhere. And I'd call that a major mistake. The Ryzen 7 7800X3D can run hot, and it definitely doesn't come with a CPU cooler. Not only should one be installed – as with all CPUs – but you'll need a reliable option to keep it cool. Going further, I'm a little surprised. AI may not get prices exactly right, but omitting a core component from a build on this list is dangerous. I'd hope most people would notice the missing part long before anything gets switched on, and only suffer a bit of delay and frustration. Because a budget is one thing, but replacing a just-bought CPU is not what anyone wants. So, beyond the otherwise fair bundle and component selection, AI is a tool to be used while acknowledging it can and still does make mistakes. Hopefully nothing that causes a CPU to spontaneously melt... but judging by the above, let's not take it for granted. If you're using AI for information on builds or CPU parts, be sure to keep your wits about you. Kevin has a broad interest and enthusiasm for consumer electronics, PCs, and all things consumer tech. And more than 17 years experience in tech journalism.

Perplexity CEO tells CNBC one metric will determine who wins the AI race - CNBC

CNBC | 2026-06-03

The companies that can provide the most economic value from the power their AI uses will ultimately command the highest valuations, Perplexity CEO Aravind Srinivas told CNBC on Wednesday. Srinivas said that whichever company can provide the "most token value per watt per user" will be the winner in the future. "Whoever is able to maximize this particular objective really will, by balancing accuracy, latency, cost, privacy and intelligence all together, they're going to win, that's what's going to win long term," Srinivas told CNBC's Elaine Yu in an interview on Wednesday. A token refers to the basic unit of data that an AI model can process. When an AI chatbot is asked to carry out a task, it breaks it down into tokens. Each token then requires energy to be processed. Srinivas' view is that whichever company can provide the best ratio of energy to economic output will be in the strongest position. "And so it might feel like some model providers are making a lot of money because their models are very expensive ... but that's short-term revenue growth," Srinivas said. Perplexity is stepping up its focus on agentic AI, a term that refers

to AI systems capable of handling more complex tasks beyond simple queries. In February, the company announced Perplexity Computer, an agent it says can execute complex tasks over long periods of time. While Perplexity develops some of its own models, its key products integrate models from other AI firms like Anthropic. A key focus for Perplexity is improving efficiency to achieve the best outcomes while minimizing energy use. To support that goal, Perplexity announced Personal Computer on Tuesday, a tool which it calls an "orchestrator." The process of orchestration involves a system that makes decisions on what the best model is to use for a particular task, how agents work together, and where the AI should process queries. Much of the actual AI processing today is done in data centers. But there is an increasing focus from AI firms on enabling these models to be processed on a device like a phone or laptop. Experts say this could reduce the power required to process AI, make it faster and more secure, since the data is not being sent to a server. Perplexity Personal Computer automatically routes the processing to where it deems best. "The data center is coming to your laptop," Srinivas said, adding that it's crucial to have an AI operating system that brings everything together in a single unified system. On Wednesday, Perplexity said that its Personal Computer product will be available on Microsoft's Windows operating system, enabling the AI to connect to apps like Word and Outlook, as well as files on a user's device. Perplexity has already launched the Personal Computer on Apple's Mac product. Srinivas said that Perplexity is focused on creating a "sustainable, durable advantage" versus competitors and that "this is an orchestration problem." "We believe that by solving that, we'll be building a pretty valuable company that has enduring, long-term advantage," Srinivas said. Perplexity faces increasing competition as rivals like OpenAI, Anthropic and Google have ramped up their focus on AI agents. Perplexity, which was last reportedly valued at \$20 billion, trails behind Anthropic and OpenAI, whose valuations have climbed to nearly \$1 trillion and just over \$850 billion, respectively. Anthropic this week confidentially filed for an initial public offering in the U.S. as investor demand for AI stocks continues. Even as Perplexity expands its products across products from Microsoft and Apple, those very same companies are developing AI agents of their own. On Tuesday, Microsoft announced new coding and reasoning models. Apple is currently working on an updated version of its digital assistant Siri based on Google's AI models. Srinivas said that while these companies will build their own AI, Perplexity's platform-agnostic approach will help it compete. "I think they absolutely will try to build their own AI systems, but we believe we're building the most versatile operating system by making it work across different models, across different chips, across different traditional operating systems, different hardware providers, different laptops," Srinivas said. "That hybrid neutral orchestration layer is what we are doing, and that allows us to balance all the different objectives simultaneously." Whenever Anthropic's models get better, Perplexity improves too because Anthropic's models are integrated into Perplexity, the CEO said. This has led Perplexity to triple its annualized revenue since the beginning of the year, "thanks to model advances that have been made by Anthropic," he told CNBC.

Perplexity AI Says 'You Can't Copyright Facts' in Defense Against CNN Copyright Suit - CNET

CNET | 2026-05-28

The television network CNN is taking aim at the artificial intelligence search engine Perplexity in a lawsuit over copyright infringement. As reported by the network's Brian Stelter, the suit, filed Thursday in a New York District Court, accuses the AI company of copying and distributing CNN's content, including over 17,000 of CNN's stories, videos, images and other published works. Though this is CNN's first legal case against an AI company, the network joins other publishers who have sued the San Francisco-based startup, including the New York Times and News Corp. According to the suit, CNN attempted to strike a licensing deal with Perplexity, but those talks didn't result in an agreement. CNN made a content licensing deal with Meta last year, where the tech giant compensates the media company for using its reporting and content to respond to queries on Meta AI. AI products regularly scrape news publications and websites to answer user questions with real-time data, accelerating the collapse in traffic and revenue to original sources. In response to the lawsuit, Jesse Dwyer, Perplexity's chief communications officer, told Stelter and other media outlets in a statement: "You can't copyright facts." The US government's Copyright Office states: "Copyright does not protect facts, ideas, systems, or methods of operation, although it may protect the way these things are expressed." CNN said in its own statement that a company valued at tens of billions of dollars shouldn't "steal from entities that create the original content Perplexity exploits" and that "commercial operators can and must pay to make use of it." A Perplexity representative didn't immediately respond to a request for comment. Perplexity is one of several companies, including OpenAI and Anthropic, that have been battling news publishers and media giants over copyright claims. (Disclosure: Ziff Davis, CNET's parent company, in 2025 filed a lawsuit against OpenAI, alleging it infringed Ziff Davis copyrights in training and operating its AI systems.) More than 100 such lawsuits have been filed. But different conclusions have been reached as to whether training AI models on copyrighted data counts as fair use, said Michael Goodyear, an associate professor at New York Law School. Considerations include how the training occurs, what AI outputs contain and whether there's any competitive harm to copyright holders. "No appellate courts have yet weighed in on the viability of these copyright infringement claims against AI companies," Goodyear said. In the CNN case, he said that Perplexity is correct that facts aren't protected by copyright, but the way CNN presents facts could be. "Even short news articles would typically qualify for copyright protection under the low bar of required originality," Goodyear said. "The question becomes whether the thousands of cases of infringement CNN describes are copying whole paragraphs verbatim, or whether they are paraphrasing or merely copying unprotectable facts." As plunging website traffic has drained billions in publisher revenue and triggered widespread media layoffs, AI firms are aggravating the crisis. According to a new report from the think tank Open Markets Institute, over the past six months, the rate of AI crawlers bypassing paywalls and blocks has nearly quadrupled, spiking

from 3.3% to 12.9%. That's partly why a number of publishers signed AI content licensing deals with tech companies to monetize content used to train AI systems. One way out for Perplexity may be to renegotiate a licensing deal with CNN. Even if Perplexity has valid legal arguments, a licensing agreement could shift from unauthorized scraping toward a formalized content partnership. However, the Open Markets Institute report says that when it comes to AI content licensing, news and content creators are trapped in a double bind. The same tech giants whose AI tools are starving websites of human traffic are now the ones gatekeeping the licensing deals meant to replace that lost ad revenue.

Perplexity is building a secret weapon to join the AI coding wars - Business Insider

Business Insider | 2026-07-07

San Francisco-based Perplexity, valued at \$20 billion, is working on 'Teammate,' an AI tool that may compete with AI coding giants like Anthropic.

Perplexity says it plans to use Nvidia's new CPU - Reuters

Reuters | 2026-07-07

Perplexity says it plans to use Nvidia's new CPU Reuters

Perplexity AI unveils hybrid local-cloud inference system at Computex 2026 - VentureBeat

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Perplexity AI, the fast-growing search startup now valued at \$20 billion, unveiled what it calls the first hybrid local-server inference orchestrator at Computex 2026 on Monday night, demonstrating software that autonomously decides — in real time and mid-task — which AI workloads stay on a user's device and which get routed to frontier models in the cloud. CEO Aravind Srinivas demonstrated the system onstage alongside Intel CEO Lip-Bu Tan during Intel's keynote address, using Perplexity's "Personal Computer" agent to process confidential deal materials. In the demonstration, local models running on Intel Core Ultra Series 3 determined which information should remain on the device and which information could be sent to cloud-based models. Srinivas said the approach balances intelligence, accuracy, privacy, and cost. The key claim is not that a model can run locally — dozens of tools already do that. It is that Perplexity's system makes the routing decision itself, task by task, without requiring the user to choose in advance. Sensitive data like financial records or health information stays on the local machine; the heavier reasoning tasks that require frontier-scale models get sent to the cloud. One task, multiple execution

locations, automatic orchestration. "No product has done this before," a Perplexity spokesperson said in an email to VentureBeat. The product is not yet available to users; according to the company, the hybrid inference feature will launch in the coming weeks. To understand why the Computex demonstration matters, it helps to trace the product arc Perplexity has been building since early this year. On February 25, Perplexity launched Computer , a multi-model AI agent that orchestrates 19 different AI models to complete complex, long-running tasks on behalf of users. The system ran entirely in the cloud, breaking goals into subtasks and routing each to whichever model — Claude, Gemini, GPT, Grok, or others — was best suited for the job. Perplexity Computer unified every current AI capability into a single system, functioning as a general-purpose digital worker that operates the same interfaces a user does. Then, in March, Perplexity introduced Personal Computer at its inaugural Ask 2026 developer conference. That product launched as a new Mac app with support for a hybrid local-cloud AI agent, which Perplexity described as a "personal orchestrator" that hybridizes local and server environments for security and productivity. Personal Computer could access the Mac's file system and native Mac apps to create and execute entire workflows, with files created in a secure sandbox and all actions auditable and reversible. What Srinivas demonstrated at Computex extends this architecture in a fundamental way. Previously, even the Personal Computer product divided labor along relatively clear lines: local file access on the device, heavy computation on Perplexity's servers. The new hybrid inference orchestrator gives the system itself the ability to reason about where each piece of a task should execute — not just which model to use, but which physical location should process it. The system reportedly asks for user permission before sending sensitive tasks to the cloud, a design choice that addresses one of the central anxieties enterprises have about agentic AI: data governance. The timing of the demonstration is not coincidental. Computex 2026 has been dominated by a single theme: on-device AI. Just hours before the Intel keynote, Nvidia CEO Jensen Huang unveiled the RTX Spark , a new Arm-based superchip that the company positions as the foundation for a new generation of AI-native Windows PCs. At full strength, the RTX Spark Superchip offers up to 20 Arm CPU cores, a Blackwell GPU with 6,144 CUDA cores, 128GB of LPDDR5X RAM, and up to 300 GB/s of memory bandwidth — enough power and memory for AI agents and 120-billion-parameter models with context lengths stretching to a million tokens. RTX Spark systems will begin arriving in the fall. Intel, not to be outdone, used its keynote to showcase Xeon 6+ processors with 288 efficiency cores built on 18A technology for the data center, and positioned its Core Ultra Series 3 as the client silicon that makes hybrid inference possible on the PC. Perplexity's hybrid orchestrator sits at the intersection of both strategies. If the system performs as advertised, it creates a direct economic incentive for users — and eventually enterprises — to invest in more powerful local silicon. The more capable the on-device chip, the more inference can run locally, reducing cloud costs and improving latency for sensitive workloads. That dynamic benefits Nvidia, Intel, and every other chipmaker competing for AI PC sockets. The implications extend well beyond chip economics. "As chips become more powerful, more intelligence moves onto a person's machine, alongside server inference for the complex tasks that still need frontier models," a Perplexity

spokesperson told VentureBeat. "Sensitive and sovereign work can stay local, which changes the need for massive country-level infrastructure." That last claim — about sovereign infrastructure — is the most provocative. Nations from the UAE to France to India have been investing billions in domestic AI compute capacity partly on the assumption that sensitive data must stay within their borders, which means building or buying access to local data centers. If meaningful inference can run on an end user's device with no data leaving the machine, the calculus changes. It does not eliminate the need for data centers, but it could soften the urgency of the buildout. Perplexity's hybrid inference play rests on the same architectural bet the company has been making all year: that the orchestration layer matters more than any individual model. For AI engineers, this signals a fundamental shift — the orchestration layer may matter more than the models themselves. The key insight is separation of concerns: the orchestration layer handles task decomposition, state management, and tool coordination, while the model layer handles specific computations. This decoupling means teams can swap models as better alternatives emerge without redesigning the entire system. Perplexity has leaned heavily into this philosophy. The company is doubling down on packaging frontier models in a consumer-friendly user experience, arguing that there is value in orchestrating multiple third-party LLMs to obtain the most cost-effective and accurate answers to queries. Models, in Perplexity's view, are specializing, not commoditizing. The hybrid inference extension takes that logic one step further. Perplexity is now orchestrating not just across models but across physical compute locations — choosing which model runs where. A lightweight local model might handle a privacy-sensitive document summarization task while a frontier cloud model tackles the complex reasoning required to analyze that summary against a broader market landscape. The orchestrator manages the handoff. This is a technically ambitious claim. Making it work reliably in production will require the orchestrator to accurately assess the complexity of each subtask, understand the sensitivity of the data involved, know the capabilities and latency characteristics of whatever local hardware the user has, and manage the state of a task that may be bouncing between environments mid-execution. It is easy to imagine edge cases where the routing logic fails, sends something sensitive to the cloud, or degrades performance by assigning a task to an underpowered local model. Perplexity says the system will be chip-agnostic, though the initial Computex demo ran on Intel silicon. The company expressed enthusiasm in its communications about the new AI chips announced at Computex this week, suggesting it intends to optimize across vendors. The hybrid inference announcement arrives at a complicated moment for Perplexity. The company has been on a remarkable growth trajectory: It secured \$200 million in new capital at a \$20 billion valuation, just two months after raising \$100 million at an \$18 billion valuation. Since its founding three years ago, the rapidly growing AI company has raised \$1.5 billion in total funding, according to PitchBook data. But the company also faces a mounting stack of legal challenges. Nine organizations have filed active suits against Perplexity for alleged copyright and trademark infringement as of May 31, 2026: CNN, the New York Times, News Corp and Dow Jones, the New York Post, the Chicago Tribune, Encyclopedia Britannica, Merriam-Webster, Reddit, and Japan's Yomiuri Shimbun. The CNN lawsuit, filed just days ago on May 28, is the most recent, accusing Perplexity of scraping more than 17,000 CNN stories,

photos, videos, and other content and using that material to train its products. Perplexity has responded with a consistent message. "You can't copyright facts," the

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Perplexity's post-trained GLM 5.2 preview pairs a cheaper open-source base with a frontier advisor—and it's already live in production. The company's post-trained GLM 5.2 preview pairs a cheaper open-source base with a frontier advisor—and it's already live in production.

References

- 1 <https://en.wikipedia.org/wiki/Perplexity>
- 2 <https://247wallst.com/investing/2026/07/12/nvidia-vs-amd-perplexity-choosing-nvidia-over-amd-tells-a-deeper-story-about-chip-dominance/>
- 3 <https://www.pcguides.com/news/gemini-perplexity-and-ai-have-picked-their-world-cup-winner-and-its-unanimous/>
- 4 <https://www.pcguides.com/news/i-asked-chatgpt-perplexity-and-gemini-to-build-me-a-gaming-pc-and-one-made-a-big-mistake/>
- 5 <https://www.cnbc.com/2026/06/03/perplexity-ceo-ai-valuations-computer-agentic.html>
- 6 <https://www.cnet.com/tech/services-and-software/perplexity-ai-cnn-copyright-suit/>
- 7 <https://www.businessinsider.com/perplexity-building-ai-coding-tool-take-on-cursor-and-openai-2026-7>
- 8 <https://www.reuters.com/business/perplexity-says-it-plans-use-nvidias-new-cpu-2026-07-07/>
- 9 <https://venturebeat.com/technology/perplexity-ai-unveils-hybrid-local-cloud-inference-system-at-computex-2026>
- 10 <https://decrypt.co/373184/perplexity-fine-tuned-chinese-ai-model-claude-opus-4-8>